* CS Stage with Current-Source Load:

In application requiring a large voltage gain in a single stage, the relationship $A_v = -g_m R_D$ suggest that we increase the load impedance of the CS stage. With a register or diode connected load, however, increasing the load resistance limits the output voltage swing.

A more practical approach is to replace the load with a a current source. The resulting circuit is shown in Fig. 1 where both transistors operate in saturation.

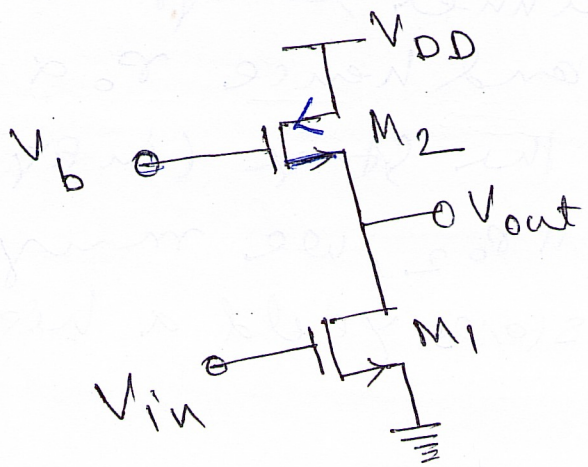


Fig. 1 CS stage with current source load

Since the total impedance seen at the output node is equal to $r_{O1} \| r_{O2}$, the gain is

$$A_V = -g_{m1}(r_{O1} \| r_{O2})$$

The key point here is that the output impedance and the minimum required $|V_{DS}|$ of M_2 are less

strongly coupled than the value and voltage drop of a resistor.

The voltage $|V_{DS2,min}| = |V_{GS2} - V_{TH2}|$ can be reduced to even a few hundred millivolts by simply increasing the width of M_2. If r_{O2} is not sufficiently high, the length and width of M_2 can be increased to achieve a smaller V while maintaining the same overdrive voltage. The penalty is the large capacitance introduced by M_2 at the output node.

We know the output impedance of MOSFET at a given drain current can be scaled by changing the channel length (i.e., to the first order, $\lambda \propto 1/L$ and hence $r_O \propto \lambda/L_D$). Since the gain of the stage (in Fig. 1) is proportional to $r_{O1} \| r_{O2}$, we may surmise that longer transistors yield a higher voltage gain.

Let us consider M_1 and M_2 separately. If L_1 is scaled by a factor α (>1), then W_1 may need to be scaled proportionally as well. This is because, for a given drain current, $V_{GS1} - V_{TH1} \propto 1/\sqrt{(W/L)_1}$, i.e., if W_1 is

not scaled, the overdrive voltage increases, limiting the output voltage swing. Also, since $g_{m1} \propto \sqrt{(W/L)_1}$, scaling up only L_1 lowers g_{m1}

In applications where these issues are unimportant, W_1 can remain constant while L_1 increases. Thus, the intrinsic gain of the transistor can be written as

$$g_{m1} r_{o1} = \sqrt{2\left(\frac{W}{L}\right)_1 \mu_n C_{ox} I_D} \cdot \frac{1}{\lambda I_D}$$

indicating that the gain increases with L because λ depends more strongly on L than g_m does. Also, note that $g_m r_o$ decreases as I_D increases.

Increasing L_2 while keeping W_2 constant increases r_{o2} and hence the voltage gain, but at the cost of higher $|V_{DS2}|$ required to maintain M_2 in saturation.

* CS Stage with Triode Load:

A Nos device operating in deep triode region behaves as a resistor and can therefore serve as the load in a CS Stage. Illustrated in Fig below (Fig. 2) such a circuit biases the gate of M_2 at a sufficiently low level. ensuring the load is in deep triode region for all output voltage swings.

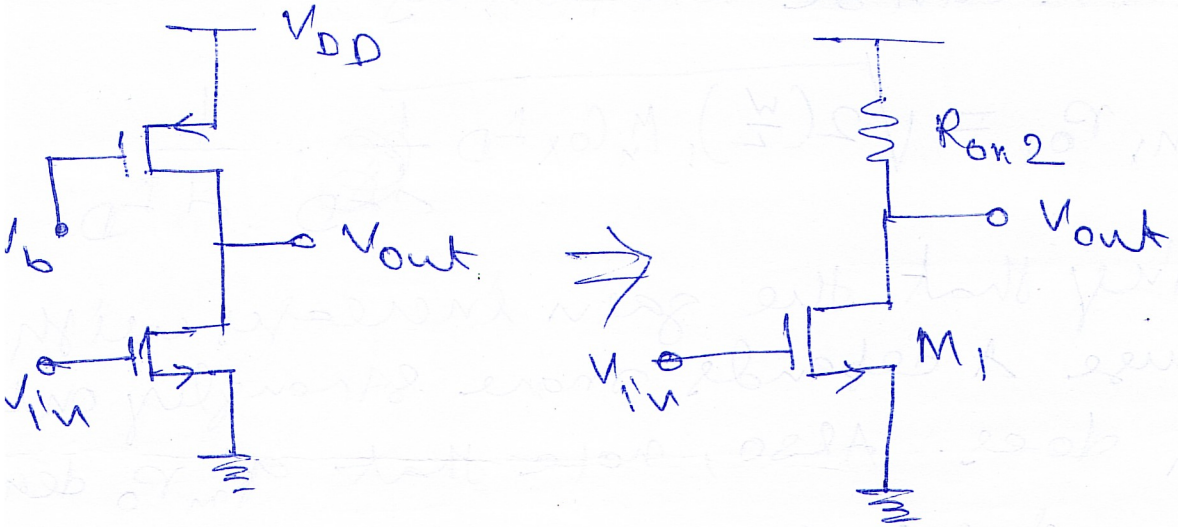


Fig. 2 CS Stage with triode load

Since $$R_{on2} = \frac{1}{\mu_p C_{ox} (W/L)_2 (V_{DD} - V_b - |V_{THP}|)}$$

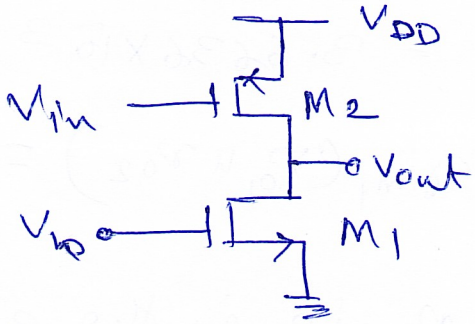
The voltage gain can be readily calculated.

The principal drawback of this circuit stems from the dependence of R_{on2} upon $\mu_p C_{ox}$, V_b, and V_{THP}. Since $\mu_p C_{ox}$ and V_{THP} vary with process

and temperature and since generating a precise value for V_b requires additional complexity, thus circuit is difficult to use. Triode loads, however consume less voltage headroom than do diode-connected devices because in CS stage triode load $V_{out,max} = V_{DD}$ whereas in CS stage diode connected load $V_{out,max} \approx V_{DD} - |V_{THP}|$.

Problems:

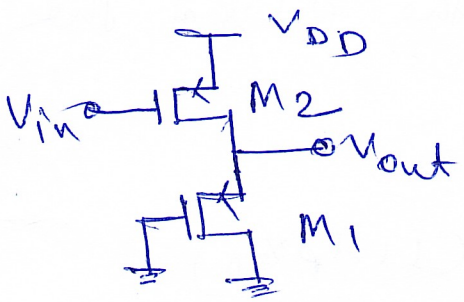
1. Fig. shows a PMOS stage using an NMOS current source load, compute the voltage gain of the circuit.



Soln: Transistor M_2 generates a small-signal current equal to $g_{m2}V_{in}$, which then flows through $r_{O1} \| r_{O2}$ producing $V_{out} = -g_{m2}V_{in}(r_{O1} \| r_{O2})$. Thus

$$\frac{V_{out}}{V_{in}} = -g_{m2}(r_{O1} \| r_{O2}).$$

2. Determine the voltage gain of the circuit shown in Fig. if $\lambda \neq 0$



Here NMOS devices changed to PMOS transistors: M_1 serves as a CS device and M_2 as a diode connected load. Thus

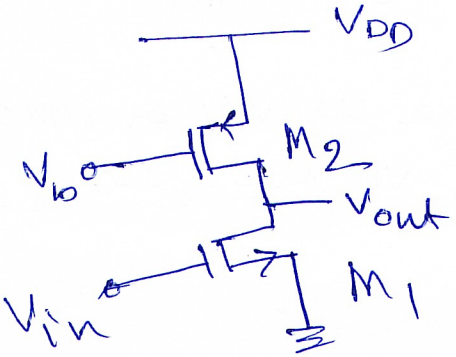
$$A_V = -g_{m2}(1/g_{m1} \| r_{O1} \| r_{O2})$$

(Fig. CS stage with diode connected PMOS device.)

Problem: On the circuit of Fig., assume $(W/L)_1 = 50/0.5$, $(W/L)_2 = 50/2$, and $I_{D1} = I_{D2} = 0.5\,mA$ when both devices are in saturation. Recall that $\lambda \propto 1/L$. $V_{DD} = 3$ volt

(a) Calculate the small-signal voltage gain.

(b) Calculate the maximum o/p voltage swing both devices are saturated.



$$r_{O1} = \frac{1}{\lambda_N I} = \frac{1}{0.1 \times 5 \times 10^{-3}} = 20\,k$$

$$r_{O2} = \frac{1}{\lambda_P I} = \frac{1}{0.2 \times 0.5 \times 10^{-3}} = 10\,k$$

$$g_{m1} = \sqrt{2 \times 1.3425 \times 10^{-4} \times 100 \times 0.5 \times 10^{-3}} = 3.6636 \times 10^{-3}$$

$$A_V = -g_{m1}(r_{O1} \| r_{O2}) = -24.40$$

If we assume that M_1 is in the edge of the triode region, then, we have:

$$V_{GS} - V_{TH1} = V_{out},\quad I_D = \frac{1}{2}\mu_n C_{ox}\left(\frac{W}{L}\right)_1 (V_{GS} - V_{TH1})^2 (1 + \lambda_n V_{DS})$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 1.3425 \times 10^{-4} \times 100\, V_{DS}^2 (1 + 0.1 V_{DS})$$

$$\rightarrow \sqrt{\frac{1}{13.425 (1 + 0.1 V_{DS})}} = V_{DS}$$

$$V_{DS\,min} = V_{O\,min} = 0.2893$$

If we assume ~~that~~ that M_2 is in the edge of the triode region, then we have:

$$I_D = \frac{1}{2}\mu_p C_{ox}\left(\frac{W}{L}\right)_2 (V_{SG} - V_{TH2})^2 (1 + \lambda_p V_{SD}),$$

$$0.5 \times 10^{-3} = \frac{1}{2} \times 3.835 \times 10^{-5} \times 25\, V_{SD}^2 (1 + \lambda_p V_{SD})$$

$$\rightarrow \sqrt{\frac{1}{0.95875 (1 + 0.03 V_{SD})}} = V_{SD} \rightarrow V_{SD\,min} = 0.9967\,V,\quad V_{O\,max} = V_{DD} - V_{SD\,min}$$

$V_{O\,max} = 2\,V$